AISD-Assignment-2 (Checkpoint-2,3)

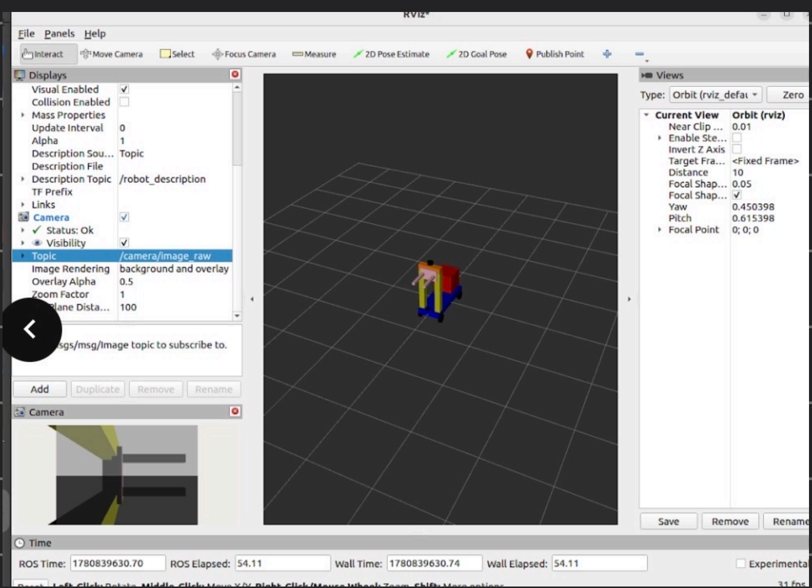
Checkpoint-2

```xml
<!--Camera link-->
<link name="camera_link">
  <visual>
    <geometry>
      <box size="0.05 0.05 0.05"/>
    </geometry>
    <material name="black">
      <color rgba="0 0 0 1"/>
    </material>
    <origin xyz="0 0 0" rpy="0 0 0"/>
  </visual>
  <collision>
    <origin xyz="0 0 0" rpy="0 0 0"/>
    <geometry>
      <box size="0.05 0.05 0.05"/>
    </geometry>
  </collision>
  <inertial>
    <mass value="0.1"/>
    <origin xyz="0 0 0" rpy="0 0 0"/>
    <inertia ixx="0.00004" ixy="0" ixz="0" iyy="0.00004" iyz="0" izz="0.00004"/>
  </inertial>
</link>

<joint name="base_camera_joint" type="fixed">
  <parent link="base_link"/>
  <child link="camera_link"/>
  <origin xyz="0.5 0 0.1" rpy="0 0 0"/>
</joint>

<gazebo reference="camera_link">
  <sensor type="camera" name="camera">
    <update_rate>30</update_rate>
    <camera>
      <horizontal_fov>1.047</horizontal_fov>
      <image>
        <width>640</width>
        <height>480</height>
        <format>R8G8B8</format>
      </image>
      <clip>
        <near>0.1</near>
        <far>100</far>
      </clip>
    </camera>
    <plugin name="gazebo_ros_lidar" filename="libgazebo_ros_laser.so">
  <topicName>/scan</topicName>
  <frameName>lidar_link</frameName>
</plugin>
  </sensor>
</gazebo>
```

Rviz2



terminal-1

Command: ros2 run robot_state_publisher robot_state_publisher --ros-args -p robot_description:="$(cat ~/AISD-ws/src/urdf-tutorials/urdf/my_robot.sensors.urdf)"

terminal-2:

command: rviz2

terminal-3:

command: ros2 run joint-state_publisher joint-state_publisher

Rviz2: ① Fixed Frame = "base_link"

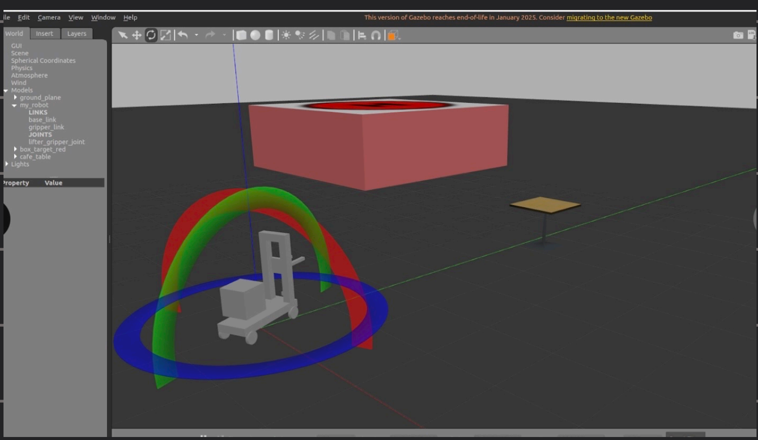
② Add → By Display Type → Robot model

choose topic as /robot_description

③ Add → By Display Type → Camera

choose topic as /camera/image-raw

Gazebo



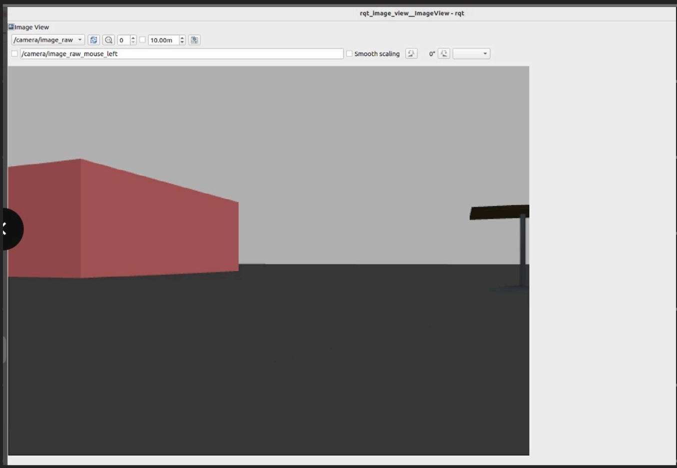
add objects → from insert

terminal-4:

Command: ros2 launch gazebo-ros gazebo.launch.py

terminal-5:

Command: ros2 run gazebo_ros spawn-entity.py -topic robot_description -entity my_robot



terminal-6:

command: ros2 run rqt-image_view rqt_image_view

command: ros2 topic hz /camera/image_raw

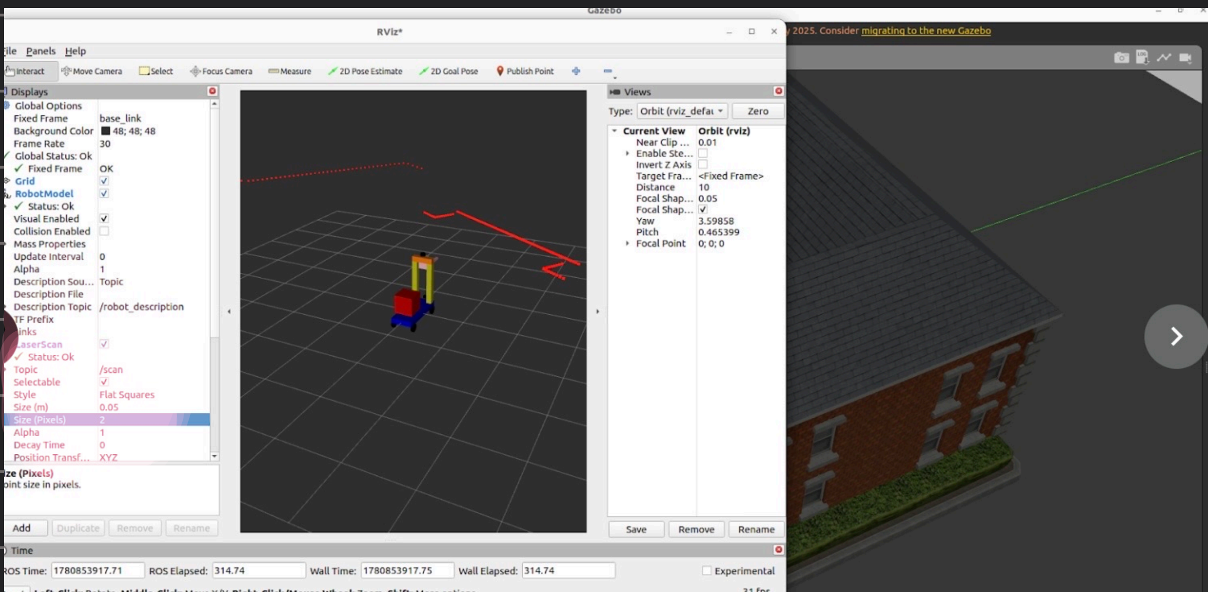
```xml
<!--LIDAR-->
<link name="lidar_link">
  <visual>
    <geometry>
      <cylinder length="0.05" radius="0.05"/>
    </geometry>
    <material name="black">
      <color rgba="0 0 0 1"/>
    </material>
    <origin xyz="0 0 0" rpy="0 0 0"/>
  </visual>
  <collision>
    <origin xyz="0 0 0" rpy="0 0 0"/>
    <geometry>
      <cylinder length="0.05" radius="0.05"/>
    </geometry>
  </collision>
  <inertial>
    <mass value="0.1"/>
    <origin xyz="0 0 0" rpy="0 0 0"/>
    <inertia ixx="0.00004" ixy="0" ixz="0" iyy="0.00004" iyz="0" izz="0.00004"/>
  </inertial>
</link>

<joint name="support_lidar_joint" type="fixed">
  <parent link="support_link"/>
  <child link="lidar_link"/>
  <origin xyz="0 0 0.1" rpy="0 0 0"/>
</joint>
```

```
<gazebo reference="lidar_link">
  <sensor type="ray" name="lidar">
    <pose>0 0 0 0 0 0</pose>
    <visualize>true</visualize>
    <update_rate>10</update_rate>
    <ray>
      <scan>
        <horizontal>
          <samples>360</samples>
          <resolution>1</resolution>
          <min_angle>-3.14159</min_angle>
          <max_angle>3.14159</max_angle>
        </horizontal>
      </scan>
      <range>
        <min>0.1</min>
        <max>10.0</max>
        <resolution>0.01</resolution>
      </range>
    </ray>
    <plugin name="gazebo_ros_lidar" filename="libgazebo_ros_ray_sensor.so">
      <ros>
        <remapping>~/out:=/scan</remapping>
      </ros>
      <output_type>sensor_msgs/LaserScan</output_type>
      <frame_name>lidar_link</frame_name>
    </plugin>
  </sensor>
</gazebo>

</robot>
```

Rviz2



terminal-1

command: ros2 run robot_state_publisher robot_state_publisher --ros-args -p robot_description:="$(cat ~/AISD_ws/src/urdf_tutorials/urdf/my_robot_sensors.urdf)"

terminal-2:

command: rviz2

terminal - 3:

command: ros2 run joint-state-publisher joint-state-publisher

Rviz 2: ① Fixed Frame = "base_link"

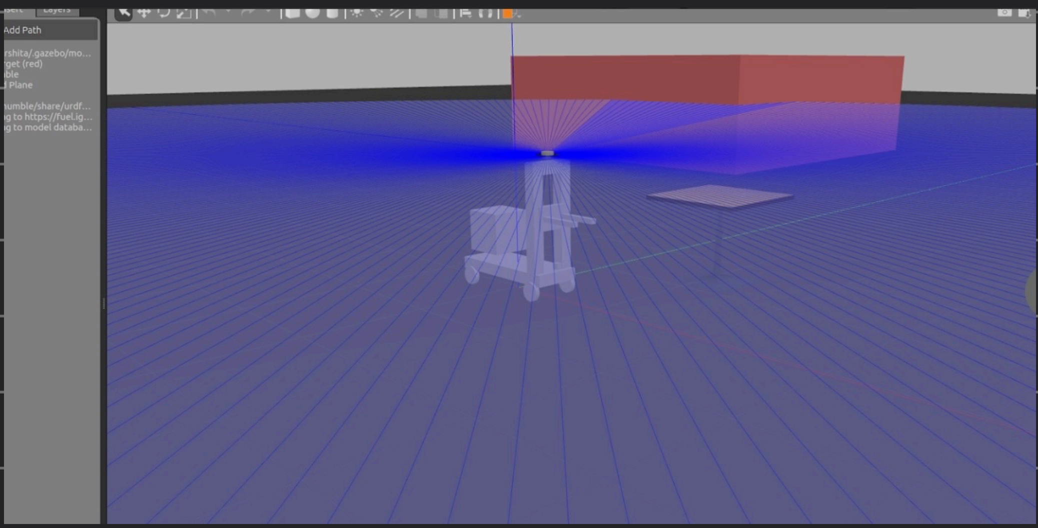
② Add → By Display Type → Robot model

choose topic as /robot_description

③ Add → By Display Type → laser Scan

↳ choose topic as /scan

Gazebo



command: ros2 topic hz /scan

terminal - 4:

Command: ros2 launch gazebo-ros gazebo.launch.py

terminal - 5:

Command: ros2 run gazebo_ros spawn-entity.py -topic robot-description -entity my_robot

Checkpoint-3

```xml

<!--Differential Drive-->
<gazebo>
  <plugin name="differential_drive_controller" filename="libgazebo_ros_diff_drive.so">
    <ros>
      <remapping>cmd_vel:=cmd_vel</remapping>
      <remapping>odom:=odom</remapping>
    </ros>
    <update_rate>50</update_rate>
    <left_joint>base_wheel_1_joint</left_joint>
    <right_joint>base_wheel_2_joint</right_joint>
    <wheel_separation>0.4</wheel_separation>
    <wheel_diameter>0.16</wheel_diameter>
    <max_wheel_acceleration>1.0</max_wheel_acceleration>
    <max_wheel_torque>20</max_wheel_torque>
    <publish_odom>true</publish_odom>
    <publish_odom_tf>true</publish_odom_tf>
    <publish_wheel_tf>true</publish_wheel_tf>
    <odometry_frame>odom</odometry_frame>
    <robot_base_frame>base_link</robot_base_frame>
  </plugin>
</gazebo>

</robot>
```

terminal-1.

Command: ros2 run robot_state_publisher robot_state_publisher -- ros-args -p robot_description:="$ (cat ~/AISD-ws/src/urdf-tutorials/urdf/my_robot_in_motion.urdf)"

terminal-4:

Command: ros2 launch gazebo_ros gazebo.launch.py

terminal-5:

Command: ros2 run gazebo_ros spawn_entity.py -topic robot_description -entity my_robot